

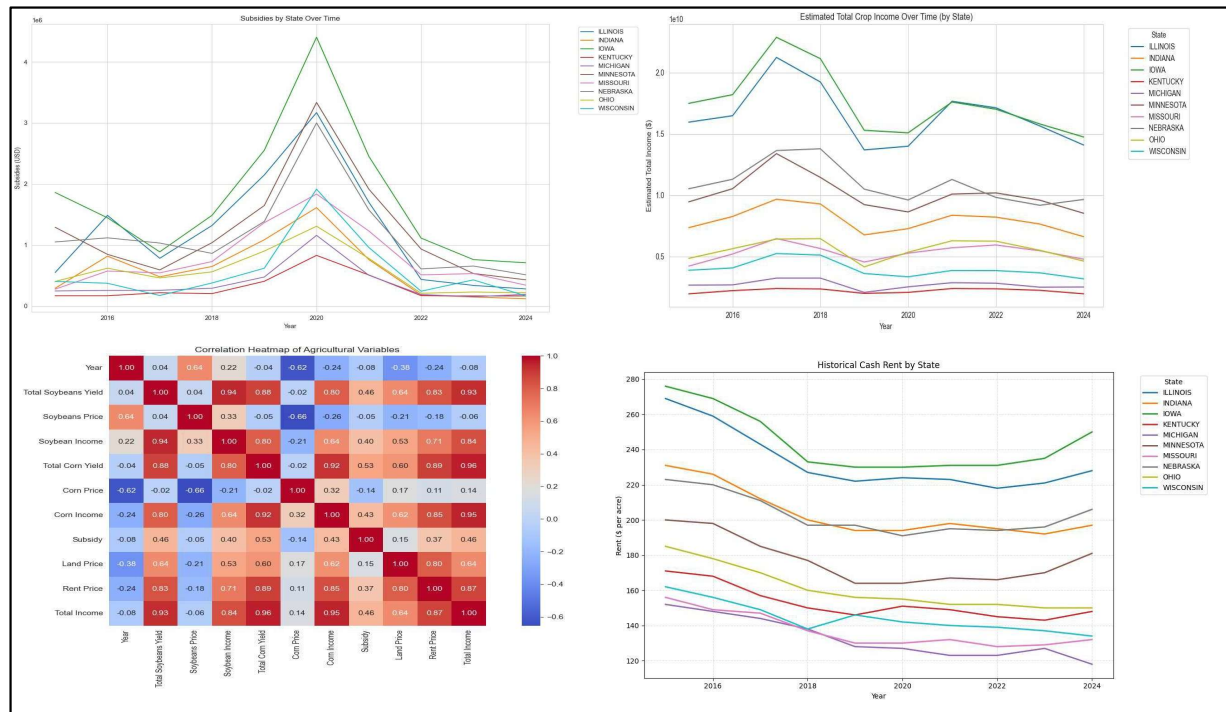
Following the Funds

Subsidy Exposure and State-Level Economic Outcomes in U.S. Row Crop Agriculture (2015–2024)

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Farm subsidies represent a major component of agricultural income in row-crop states, but their relationship with farm performance is difficult to assess due to fragmented and aggregated data. This project examines how federal subsidy payments relate to farm income, yields, and land-based economic indicators across U.S. states from 2015–2024.

The dataset was structured as a state-year panel and cleaned for consistency across variables. Analysis includes time-series line charts to show trends, bar charts for cross-state comparisons, and correlation heat maps to identify relationships between variables. These methods were chosen to emphasize temporal dynamics and inter-variable relationships within a structured panel dataset.



The results show that agricultural subsidies are substantially more volatile than farm income, with pronounced variation across states and a notable spike around 2020. In contrast, total crop income exhibits smoother year-to-year changes, with most states following similar trends aside from the 2020 disruption. The divergence in volatility suggests that subsidy payments respond to policy and market shocks, while income is primarily driven by underlying production conditions and commodity prices. Overall, visual evidence indicates weak direct co-movement between subsidies and income at the aggregate state level.

The analysis uses USDA QuickStats state-level survey data from 2015–2024. Key variables include subsidy payments, corn and soybean yields, commodity prices, land values, and cash rents. Total crop income is constructed as planted acres multiplied by yield and price. Subsidy measures represent aggregated annual state payments and do not include crop insurance subsidies.